

- 4 (a) when a wave (front) passes by/incident on an edge/slit M1
 wave bends/spreads (into the geometrical shadow) A1 [2]

(b) $\tan \theta = \frac{38}{165}$

$\theta = 13^\circ$ C1

$d \sin \theta = n\lambda$ C1

$d = 2.82 \times 10^{-6}$ C1

number = $(1/d =) 3.6 \times 10^5$ A1 [4]

- (c) P remains in same position B1
 X and Y rotate through 90° B1 [2]

- (d) *either* screen not parallel to grating
or grating not normal to (incident) light B1 [1]

- 5 (a) region/area where a charge experiences a force B1 [1]